Self-Justification Ex Machina

jpt4*

In memory of Dan E. Willard (1948–2023)

Abstract

Sebastian Oberhoff's essay *Incompleteness Ex Machina* proves Gödel's incompleteness theorems twice, once with sentences and once with programs, and concludes that Gödel's work divides into two parts: teaching arithmetic to compute, and the incompleteness theorems, which are victory laps. We look closely at the one step of the second theorem that Oberhoff takes "with some faith". It asks a formal system to believe that whenever it can watch a program halt, it can also write down a transcript of the run, and transcripts are bigger than what they record. That belief is a claim about how fast numbers grow. Dan Willard's self-justifying axiom systems are what remains when the belief is withheld. They watch computations as well as any arithmetic does, they are incomplete, and they take their own consistency as an axiom and are right. We retell Willard's consistency argument as a program in Oberhoff's style, show that the faith step cannot be proved in such a system (Oberhoff's own argument, run backwards), and locate where the faith comes back: at multiplication for proofs that never reuse a lemma, and already at counting for proofs that do. Gödel's work, rightly viewed, splits into three parts, not two.

> Another way to motivate the axiom system IS is to consider a philosopher's response to the question of whether multiplication is a function. His first answer would likely be "of course yes". But if one then asked the philosopher to construct a sequence of natural numbers $a_0, a_1, a_2, a_3, \ldots$ where $a_0 \geq 2$ and $a_{i+1} = (a_i)^2$, a hedge would quickly follow because a_n requires at least 2^n bits. The expected response would then be that "multiplication is a function in theory but not always in practice". — Dan Willard, SUNY Albany TR 93-10, p. 6

1 Victory Laps

Oberhoff's essay [Oberhoff 2018] does something deceptively simple. It writes out Gödel's first incompleteness theorem, Rosser's improvement of it, and the second incompleteness theorem as explicit case analyses. Then it replaces Gödel's sentence G ("I am not provable") by $\ddot{\mathsf{G}}$ = «P(P) doesn't halt», where P is the program

```
P(A) {
  for s ∈ all possible strings {
    if s is a proof in 𝓕 that A(A) doesn't halt {
      halt;
    }
  }
}
```

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and observes that each case analysis can be copied almost word for word. The punchline (his §5) is that $\ddot{\mathsf{G}}$ is G wearing Clark Kent's glasses: $\mathsf{P}(\mathsf{P})$ is searching $\mathcal{F}$'s proofs for a proof of $\ddot{\mathsf{G}}$, so $\ddot{\mathsf{G}}$ says "I am not provable". His moral (§6) is a refactoring. The hard part of Gödel's work was teaching arithmetic to compute, packaged as

> **Theorem 7** (Turing Completeness of Elementary Arithmetic). *Every formal system capable of reasoning about elementary arithmetic can also reason about algorithms,*

which Oberhoff leaves "to smarter people". The incompleteness theorems themselves are victory laps.

For the first incompleteness theorem we accept this completely. This essay is about the second. Its proof in [Oberhoff 2018] contains exactly one step that Oberhoff marks as taken on trust (Theorem 6, third step):

> Now we simply observe (with some faith) that there wasn't actually anything in the reasoning we used to prove this that we couldn't carry out just as well in $\mathcal{F}$.

For the systems Oberhoff has in mind, that faith is well placed. But some formal systems can reason about algorithms in every sense the first theorem needs, and for them the faith is misplaced and the conclusion of Theorem 6 is false: they prove their own consistency, and they are consistent. Dan Willard constructed them between 1993 and 2021 [Willard TR 93-10, Willard 2001, Willard 2005, Willard 2006a, Willard 2020]. (We cite Willard's papers by year, as in the bibliography: "2002a" is [Willard 2002a].) This essay tells their story in Oberhoff's language. Our headline is that the faith step is not part of the victory lap. It is a third part of Gödel's work, and it has a price.

2 Where the Faith Lives

Here is the first half of Oberhoff's proof that $\ddot{\mathsf{G}}$ is undecidable (his §4, a second proof of his Theorem 4):

Suppose $\ddot{\mathsf{G}}$ is provable:
Or $\neg\ddot{\mathsf{G}}$ isn't provable:

- $\Rightarrow$ Because $\ddot{\mathsf{G}}$ is provable $\mathsf{P}(\mathsf{P})$ will eventually find such a proof after some sequence of steps which we can write out in $\mathcal{F}$.
- $\Rightarrow$ Because this sequence leads to a terminal state this gives a proof of «$\mathsf{P}(\mathsf{P})$ halts» $= \neg\ddot{\mathsf{G}}$. $\lightning$

Theorem 6 asks $\mathcal{F}$ to run this argument on itself. Written as statements that $\mathcal{F}$ must prove, the argument has three steps.

(1) «If $\mathsf{P}(\mathsf{P})$ halts, then $\ddot{\mathsf{G}}$ is provable.» This is cheap. P halts only after it has found a proof of $\ddot{\mathsf{G}}$. If we formalize «$\mathsf{P}(\mathsf{P})$ halts» as «P's search succeeds», which is how arithmetic usually formalizes it, the two sentences are the same sentence.[1]

(2) «If $\ddot{\mathsf{G}}$ is provable, then $\neg\ddot{\mathsf{G}}$ is provable.» This is the step "$\mathsf{P}(\mathsf{P})$ will eventually find such a proof after some sequence of steps which we can write out in $\mathcal{F}$". Inside $\mathcal{F}$ it reads: *for every proof s of $\ddot{\mathsf{G}}$ there is a proof of «$\mathsf{P}(\mathsf{P})$ halts»*, namely the proof that exhibits s and checks it.

[1] If halting is formalized through execution traces instead, even this step costs something: given a proof s, $\mathcal{F}$ must believe that the trace of the search up to s exists. Willard always works with proof predicates directly.

(3) «If $\ddot{\mathsf{G}}$ and $\neg\ddot{\mathsf{G}}$ are both provable, then $\mathcal{F}$ is inconsistent.» This is free if "consistent" means that no sentence is provable together with its negation. If it means that there is no proof of $0 = 1$, then $\mathcal{F}$ must also believe that two proofs can be combined into a third. That is cheap with modus ponens and expensive without it (§7).

Chaining (1)–(3) gives $\mathcal{F} \vdash$ «$\neg\ddot{\mathsf{G}} \Rightarrow \mathcal{F}$ is inconsistent», that is, $\mathcal{F} \vdash$ «$\mathcal{F}$ is consistent $\Rightarrow \ddot{\mathsf{G}}$», which is what Theorem 6 needs.

The only step that asks $\mathcal{F}$ to know something substantial about itself is (2). Look at its shape. In the first incompleteness theorem, "which we can write out in $\mathcal{F}$" was a fact *about* $\mathcal{F}$, established by us from outside: we are handed a halting run and we produce the proof. In the second theorem it has to be a theorem *of* $\mathcal{F}$, stated for every s at once. The two deserve separate names.

Definition A (Watching)**.** *$\mathcal{F}$ watches* algorithms if, whenever a program halts, $\mathcal{F}$ proves that it halts.

Definition B (Knowing that one watches)**.** *$\mathcal{F}$ knows that it watches* if $\mathcal{F}$ proves, for programs A, «if A halts, then $\mathcal{F}$ proves that A halts».

Watching is external Σ_1-completeness. Knowing that one watches is formalized Σ_1-completeness, in computational dress; the third Hilbert–Bernays–Löb derivability condition [Hilbert–Bernays 1939, Löb 1955], which the classical proof of the second theorem uses, is the special case in which the program is a proof search. Here is what each result of [Oberhoff 2018] uses:

Result in [Oberhoff 2018]	Assumes about $\mathcal{F}$	Watches	Knows it watches
Thm 4 (G1 via halting)	honest; complete (for contradiction)	—	—
§4, $\ddot{\mathsf{G}}$ undecidable	consistent; honest in case 2	case 1	—
Thm 5 (Rosser, via B)	consistent	both cases	—
Thm 6 (G2)	consistent	yes	step (2)

Theorem 4 needs only that $\mathcal{F}$ can *state* halting facts honestly; the search does the rest. Rosser's case 2 also uses the fact that a program which has entered `while true {}` never leaves it, a Π_1 fact that Oberhoff's footnote 12 says every respectable system can prove. Apart from that, every result in the essay except the second incompleteness theorem uses watching and nothing more. That is no accident of presentation. The classical proof of the second theorem runs through the derivability conditions; the computational retelling shows what the third condition *says*. It says that transcripts exist.

3 Transcripts Are Bigger Than Runs

Why should knowing be harder than watching? Because of a size mismatch. A proof of «$\mathsf{P}(\mathsf{P})$ halts» has to contain s, or at least name it, and check it step by step. Written out, the check is bigger than s. (Replaying P's whole search up to s would be far bigger still.) So step (2) asks $\mathcal{F}$ to prove a statement of the form "for every s there exists t", where t is a number larger than s, obtained from s uniformly. That is a totality claim: $\mathcal{F}$ must believe that the function taking a proof to its checked transcript is defined everywhere. A totality claim about a function that grows is a claim about how big numbers get.

Most arithmetics accept it without blinking. Willard noticed that one can refuse it without refusing to compute. His first statement of the idea continues this essay's epigraph (TR 93-10, p. 7, verbatim):

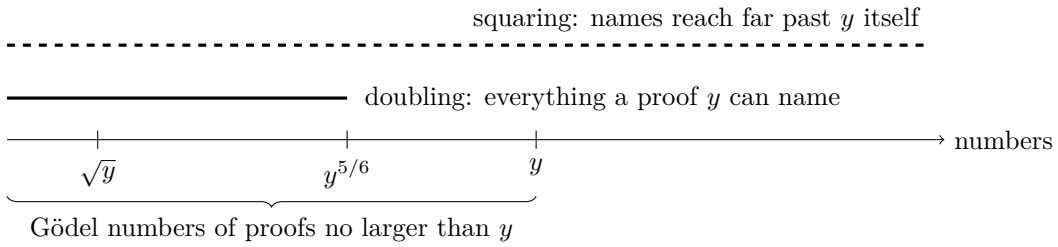


Figure 1: The horizon of a proof (schematic). In Willard's 2001 accounting, a tableau proof y mentions only constants below $\sqrt{y}$, and with doubling as its fastest operation it can enlarge them by at most $y^{1/3}$ along the branches that matter, so it names nothing beyond $y^{5/6} < y$ (§5). With squaring, a few symbols name numbers far beyond y.

> This means there will always be adequate memory for writing b_n's binary encoding when there is sufficient memory for storing a semantic tableaux proof of its existence (whereas the same is plainly not true *when multiplication is assumed to be a function*).
>
> For instance, multiplication allows one to establish the existence of a number a_{300} (in a proof with approximately 300 lines), whose binary encoding requires *more digits* than the number of atoms in the universe.

Here $b_{i+1} = b_i + b_i$ and $a_{i+1} = (a_i)^2$. The arithmetic is simple. Start both sequences at 2:

$$x_n = 2^{n+1} \text{ (doubling)}, \qquad y_n = 2^{2^n} \text{ (squaring)}, \qquad \log_2 x_n = n + 1, \qquad \log_2 y_n = 2^n.$$

A proof that builds x_n by n doublings has about n steps, and x_n has about n binary digits: the number fits in the memory its proof occupies. A proof that builds y_n by n squarings also has about n steps, but y_n has 2^n digits. Willard's later statement is sharper. It compares the Gödel number of a proof with the integer the proof shows to exist (Willard 2006a, Remark 3, p. 24, verbatim):

> In particular, every proof of the existence of an integer N using the additive convention will have a larger Gödel number than N. In contrast, the multiplicative convention allows for the Gödel number of such proofs to have a sharply smaller magnitude than N (when for instance $N = 2^{2^k}$ and k is larger than some fixed constant). In essence, this difference in magnitude is the reason the Second Incompleteness Theorem will generalize under the multiplicative naming paradigm — while Section 4 showed boundary-case exceptions to it exist under the additive naming convention.

Put the two observations together. Step (2) asks for transcripts, which are bigger than what they transcribe. In a system whose proofs cannot establish the existence of numbers bigger than themselves, step (2) is in trouble. For now this is a heuristic; §6 turns it into a theorem, and the sections between describe systems built so that it holds.

4 A Machine That Says "I Am Consistent"

Oberhoff's moral for the second theorem is "if you declare yourself a very stable genius, you're not". Let us build the genius with his tools and watch it fall. Fix an honest base system $\mathcal{F}_0$ that can reason about algorithms, and consider

```
C(A) {
  for s ∈ all possible strings {
    if s is a refutation of F_0 + «A(A) doesn't halt» {
      halt;
    }
  }
}
```

A *refutation* of a system is a proof of $0 = 1$ from it. (We picked C because it is a circle that never quite closes; §5 explains the joke.) Let

$$\ddot{\mathsf{W}} = \text{«}\mathsf{C}(\mathsf{C}) \text{ doesn't halt»}, \qquad \mathcal{F} = \mathcal{F}_0 + \ddot{\mathsf{W}}.$$

Just as $\ddot{\mathsf{G}}$ says "I am not provable", $\ddot{\mathsf{W}}$ says "nothing refutes $\mathcal{F}_0$ together with me". In other words, $\ddot{\mathsf{W}}$ says that $\mathcal{F}$ is consistent. So $\mathcal{F}$ proves its own consistency in one line, because the claim is an axiom. This is the self-referential axiom of Willard's systems, built (like $\ddot{\mathsf{G}}$) by the recursion theorem. Willard (2016, Definition 3.4 and Example 3.5) remarks that constructing it is "relatively easy, via an application of the Fixed Point Theorem, but this sentence is ironically typically useless!"

The uselessness takes one line to prove.

Proposition C (The genius falls)**.** *Let $\mathcal{F}_0$ be Peano Arithmetic, or any other system strong enough that every effectively axiomatized extension of it knows that it watches. Then $\mathcal{F} = \mathcal{F}_0 + \ddot{\mathsf{W}}$ is inconsistent, and $\mathsf{C}(\mathsf{C})$ halts.*

Proof. Suppose $\mathcal{F}$ is consistent.

$\Rightarrow$ $\mathcal{F}$ proves its own consistency, since $\ddot{\mathsf{W}}$ is one of its axioms.

$\Rightarrow$ $\mathcal{F}$ satisfies every hypothesis of Oberhoff's Theorem 6, including the faith step.

$\Rightarrow$ So $\mathcal{F}$ cannot prove its own consistency. $\lightning$

So $\mathcal{F}$ is inconsistent. A refutation exists (it is Theorem 6's argument, written out in $\mathcal{F}$), and $\mathsf{C}(\mathsf{C})$ finds it. □

Willard records that "Kleene, Rogers and Jeroslow [...] each warned their readers that most axiom systems, similar to α^*, were useless on account of their inconsistency, although they were technically well-defined" (2005, §1). With faith, the genius falls on the first day.

5 Willard's Machine

Willard keeps C and $\ddot{\mathsf{W}}$ and changes $\mathcal{F}_0$. In the version that is easiest to explain, the tableaux line of 2001–2005, $\mathcal{F}_0$ has three groups of axioms and one restriction on proofs (codified statement §5.1):

- **Names.** Constants for 0, 1, 2, with addition total. In the 2005 version, addition and $Double(x) = x + x$ are function symbols, so every number n has a name of $O(\log n)$ symbols. Multiplication is *not* a function. It is a three-place relation $M(x, y, z)$, and $\mathcal{F}_0$ does not prove $\forall x \forall y \exists z\, M(x, y, z)$. (Multiplication by any *fixed* constant remains total; what is withheld is uniform multiplication.)
- **Watching.** A finite set of Π_1 axioms that proves every true bounded (Δ_0) sentence and gives $=$ and $\leq$ their usual properties. "This finite string is a halting run of A" is a bounded sentence, so $\mathcal{F}_0$ watches every halting program.

- **Inheritance.** For each Π_1 sentence Φ, the axiom «for every y, if y is a proof of Φ in A, then Φ», for a chosen base theory A such as Peano Arithmetic. Because $\mathcal{F}_0$ watches, it can verify any particular A-proof, and so it proves every Π_1 theorem of A. That includes every fact of the form "this program never halts" that A knows, footnote 12's `while true {}` among them.
- **Proofs without lemmas.** The proof system is semantic tableaux: proof by refutation tree, with no cut rule and no modus ponens. (Willard 2005 later admits lemmas, but only Π_1 and Σ_1 ones: subroutine calls with a very simple interface.)

Theorem D (Willard)**.** *If the* Π_1 *theorems of* A *are true, then* $\mathcal{F} = \mathcal{F}_0 + \ddot{\mathsf{W}}$ *is consistent.*

This is Willard 2001, Theorem 4.3 (p. 20), for the system IS(A), whose $\ddot{\mathsf{W}}$ forbids proofs of $0 = 1$; and Willard 2005, Theorems 3 and 5 (pp. 27, 29), for IS$_D(A)$, whose $\ddot{\mathsf{W}}$ forbids proving any Π_1 sentence together with its negation, under tableaux and under tableaux with Π_1/Σ_1 lemmas respectively. Both proofs are complete (status `full`). Like every consistency proof, this is a theorem of the metatheory, proved under a hypothesis (here, that A's Π_1 theorems are true, and that A's proofs are encoded in the standard bounded way); it is a relative consistency result.

By Oberhoff's standards, then, $\mathcal{F}$ is a formal system capable of reasoning about algorithms. It watches, and it proves every Π_1 theorem of Peano Arithmetic. It is honest: its base axioms are true, its added axiom $\ddot{\mathsf{W}}$ is Π_1, and a consistent extension of a true, Σ_1-complete base by Π_1 axioms proves only truths.[2] It is incomplete: Oberhoff's Theorem 5 goes through verbatim, because Rosser's program B is watched in both cases and the loop fact is a Π_1 theorem of Peano Arithmetic. Willard said as much in his first technical report: IS(A) is "a class of structures, lying halfway between Peano Arithmetic and Presburger Arithmetic, which satisfy Godel's First but not Second Incompleteness Theorem", because the predicate $\mathit{Turing}(x, y, z)$ is expressible in it (TR 93-10, p. 6). And yet $\mathcal{F}$ proves its own consistency.

Here is why it is consistent, in Oberhoff's format. This is Willard's own sketch (2001, Remark 4.4, pp. 20–21), put in computational words.

Suppose $\mathcal{F}$ is inconsistent:

$\Rightarrow$ Then $\mathsf{C}(\mathsf{C})$ halts. There exists some shortest refutation y.

$\Rightarrow$ y is a tableau, a tree in which every branch closes. Walk down y, always choosing a branch on which everything said so far is true of the numbers named so far. Everything $\mathcal{F}_0$ says is true, so only $\ddot{\mathsf{W}}$ can stop the walk.

$\Rightarrow$ $\ddot{\mathsf{W}}$ says «no s is a refutation of $\mathcal{F}$». A tableau uses a universal axiom only by instantiating it at a term the branch has already built. So $\ddot{\mathsf{W}}$ can stop the walk only at a number named on the branch that actually bounds a refutation of $\mathcal{F}$.

$\Rightarrow$ But y can name only numbers smaller than itself. Its fastest operation is doubling, every doubling costs symbols, and y has too few symbols to double its way past its own Gödel number.

$\Rightarrow$ So $\mathcal{F}$ has a refutation smaller than y, which was the shortest one. $\lightning$

Hence $\mathcal{F}$ is consistent. Since $\ddot{\mathsf{W}}$ says exactly that, $\ddot{\mathsf{W}}$ is true, and $\mathcal{F}$ is honest.

Willard compresses the argument into one sentence (2001, p. 20, verbatim): "any proof y of $0 = 1$ from IS(A) will be automatically associated with an ordered pair (p, u) where $p \leq \text{VALUE}(u) < y$ and p represents ANOTHER proof of $0 = 1$". He calls this a *vacuous truth*, an implication whose

[2] A false Π_1 axiom would have a true Σ_1 negation, which the base proves, so the extension would be inconsistent. See the repository note `docs/log/2026-09-24-soundness-of-willard-sjas.md`.

hypothesis is never satisfied, and notes that once it is established "the remainder of the proof of Theorem 4.3 is fairly routine and easy". All the work is in the fourth arrow. The formal proof replaces the walk by an eleven-case induction over the tableau rules (Lemma 4.8, p. 23, `full`). The induction shows that the least refutation contains an "affirmative" branch, and such a branch cannot close (Lemma 4.6, p. 22, `full`). The naming bound comes from Lemma 4.1 (a constant mentioned in y has value below $\sqrt{y}$, p. 17, `full`) and Lemma 4.2 (along such a branch every parameter has value below y, p. 19, `sketch`). In numbers: there are fewer than $\frac{1}{3}\log_2 y$ doublings, so values reach at most $y^{1/2} \cdot y^{1/3} = y^{5/6} < y$. That is Figure 1.

Three remarks.

The proof is a program. The walk is an algorithm, later named *Probe* (Willard 2005, Lemma 1, p. 23, `full`). It takes any purported refutation and returns an open branch, together with values for its variables that make every sentence on the branch true. Willard's consistency proof is therefore a machine that defeats each refutation by running it, in a universe barely larger than the refutation itself. That is self-justification ex machina in the most literal sense. (And it is why we named the searcher C: it hunts for a tree all of whose branches close, and the walk always finds one that stays open.) $\mathcal{F}$ cannot prove that the machine works; that is the external part. $\mathcal{F}$ has only the axiom.

The encoding is part of the argument. The horizon is measured in Gödel numbers, so it depends on how proofs are written. In Willard's 2001 accounting, every addition can double a value, and the margin absorbs this only because writing down the sentence that introduces the sum "requires substantially more than six bits" (2001, p. 28). In the 2005 version the coding density 1/5 (Eq. 20, six-bit bytes) enters the bound directly. A proof notation compact enough to let a proof name numbers larger than itself would break the argument. Encoding choices are not bookkeeping here; they drive self-justification. (Willard notes that on the Hilbert line his results survive uncompressed encodings, though less meaningfully. Density is load-bearing for the tableaux line in particular.)

There are two horizons. The 2005 version, with $\ddot{\mathsf{W}}$ forbidding a Π_1 sentence Ψ and its negation, replaces the walk with a quantitative statement. Take a minimal pair of proofs of Ψ and of $\neg\Psi$, and let m be the larger. Below m, $\ddot{\mathsf{W}}$ is true. Willard's *θ-compactification* says that a proof of length p can stretch a sentence's reach by at most a factor p^θ, and tableaux achieve $\theta = 1/4$ (2005, Definition 5 and Theorem 2, pp. 19, 26, `full`). So the proof of the universal sentence Ψ certifies it out to $(m-1)/m^{1/4}$, while the proof of the existential $\neg\Psi$ finds its counterexample below $m^{1/4}$. For $m \geq 3$ the counterexample lies where Ψ has been certified. $\lightning$ (Theorem 1, p. 19, `full`. The printed proof treats the closing inequality as immediate; it needs $m \geq 3$, which always holds for the Gödel number of a proof.) A proof of "always" guarantees its claim out to its horizon, and a proof of "sometimes" finds its witness inside its horizon. When horizons are short compared with the proofs, the two collide in the region where every axiom is true.

6 The Faith Is Not Just Missing

One might think Willard's systems are self-justifying merely because nobody has managed to prove step (2) in them. In fact they cannot prove it. The reason is Oberhoff's Theorem 6, run backwards.

Proposition E. *Let $\mathcal{F}$ be consistent and effectively axiomatized. Suppose $\mathcal{F}$ watches, proves that no Π_1 sentence is provable in $\mathcal{F}$ together with its negation, and proves the fixed-point equivalence $\ddot{\mathsf{G}} \leftrightarrow \neg\mathrm{Prov}_{\mathcal{F}}(\ulcorner\ddot{\mathsf{G}}\urcorner)$ for its own $\ddot{\mathsf{G}}$. Then $\mathcal{F}$ does not prove step (2):*

$$\mathcal{F} \nvdash \mathrm{Prov}_{\mathcal{F}}(\ulcorner\ddot{\mathsf{G}}\urcorner) \Rightarrow \mathrm{Prov}_{\mathcal{F}}(\ulcorner\neg\ddot{\mathsf{G}}\urcorner).$$

Proof. Suppose $\mathcal{F}$ proves step (2).

$\Rightarrow$ $\mathcal{F}$ proves step (1), which is half of the fixed-point equivalence. $\mathcal{F}$ proves step (3), because $\ddot{\mathsf{G}}$ is a Π_1 sentence and step (3) is an instance of $\mathcal{F}$'s self-consistency.

$\Rightarrow$ Chaining (1)–(3): $\mathcal{F} \vdash \neg\ddot{\mathsf{G}} \Rightarrow \bot$, so $\mathcal{F} \vdash \ddot{\mathsf{G}}$. (Theorems are closed under logical consequence for every complete proof system, with or without cut [Willard 2020].)

$\Rightarrow$ So $\mathsf{P}(\mathsf{P})$ finds this proof and halts.

$\Rightarrow$ $\mathcal{F}$ watches $\mathsf{P}(\mathsf{P})$ halt: $\mathcal{F} \vdash \neg\ddot{\mathsf{G}}$. So $\mathcal{F}$ is inconsistent. $\lightning$

□

Willard's $\mathrm{IS}_D(A)$ satisfies every hypothesis except possibly the last (2005, Eq. 6 and Theorems 3 and 5). The last, the fixed-point equivalence, is assumed here as for Willard's own self-referential axiom; for $A \supseteq$ Peano Arithmetic, the uniqueness facts about the substitution function that it needs are true Π_1 theorems of A and are therefore inherited. Hence *a consistent system that proves its own consistency cannot know that it can watch its own proof-searcher halt.* Step (2) is in fact true in such a system, since its antecedent is false ($\ddot{\mathsf{G}}$ is unprovable), so it is a true sentence that the system cannot prove. Oberhoff's faith step is the system's own blind spot. For systems whose $\ddot{\mathsf{W}}$ only forbids proofs of $0 = 1$, such as the 2001 $\mathrm{IS}(A)$, step (3) also costs something (combining two proofs into a refutation needs a cut), so the faith fails in at least one of two places.

The proposition is a contrapositive and claims no novelty. Willard's own 1994 framing already sets the alternatives out: "every axiom system either fails to prove its consistency or fails to meet the full Hilbert-Bernays conditions" (TR 93-10, p. 6). What the computational reading adds is *which* condition fails, at *which* sentence, and the reason it fails. What Willard adds, and what no contrapositive can supply, is that systems on the far side of the alternative *exist*, and are strong enough to prove every Π_1 theorem of Peano Arithmetic.

7 The Price of Faith

If withholding faith is what lets $\ddot{\mathsf{W}}$ survive, the second incompleteness theorem should return as soon as the system has enough growth to know that transcripts exist. Willard spent the second half of his career proving that it does, and pricing the faith exactly. The price depends on the proof system. The difference is the difference between a program that may call subroutines and one that may not.

Squaring

Give the tableaux system multiplication as a total function. Then a proof can introduce $u_0 = 2$, $u_{i+1} = u_i \cdot u_i$, and so reach $u_n = 2^{2^n}$, in $O(n)$ steps. Proofs can now establish the existence of numbers far larger than themselves, so transcripts fit, and the walk of §5 fails at its fourth arrow. Willard located the failure precisely (2000, §5, quoted in the codified statement §5.3):

> Many readers will smile with amusement when they learn the true reason that the Semantic Tableaux version of the Second Incompleteness Theorem breaks down when Multiplication is changed from a total function into a 3-way relation. It is essentially that Lemma 8 and its short proof then become no longer valid. ... Without it, our proof of the Semantic Tableaux version of the Second Incompleteness Theorem collapses entirely!

Lemma 8 is the squaring construction. With it, the second theorem comes back. No consistent finite extension of $Q + V$, for an explicit Π_1 sentence V, proves its own tableaux consistency (2002a, Theorem 3.5, p. 11, `full`). The same holds for every consistent extension of the textbook and

Wilkie–Paris axiomatizations of $I\Sigma_0$ (2007, Theorem 6, p. 29, `full`), and, under a technical condition on how a system recognizes its own axioms, for extensions of $I\Sigma_0$ generally (2002a, Theorem 6.4, p. 25, `sketch`; this answers a 1981 question of Paris and Wilkie). From the positive side, the compactification engine becomes "impossible to satisfy" once multiplication is admitted as a ninth function symbol (2005, Remark 4, p. 21).[3]

Subroutines

A Hilbert-style proof can prove a lemma once and use it many times. Modus ponens is a subroutine call. A tableau cannot do this; every use must be inlined. Inlining can be astronomically expensive: cut elimination preserves theorems, but the cut-free proof "can certainly be super-exponentially longer" (2002a, Theorem 2.2, p. 7), and Boolos gave a famous small example of the blowup [Boolos 1984]. With subroutines, a little growth is amplified into a lot. Willard saw this in 1994 (TR 93-10, p. 7, verbatim):

> These metaphors lose meaning when cuts are present because Addition can then be employed to *implicitly define* a sequence that grows actually faster than $a_0, a_1, a_2, a_3 \ldots$. Solovay's proof that IS(A) is incompatible with deductive cuts [So94] rests on this fast growth property. (Actually, IS(A) can support a cut rule for Δ_0 sentences, and Solovay's theorem relies on the cut rule for sentences with unbounded quantifiers.)

In outline, the mechanism is the classical shortening of definable cuts. With lemmas, one defines a set of numbers and proves once that it is closed under successor; then one proves, by a short proof reusing that lemma, that a smaller set is closed under addition, and then that a smaller set still is closed under multiplication. On the last set the system, roughly speaking, knows that transcripts exist, and Gödel's argument runs there: Pudlák showed that a consistent extension of Robinson's Q cannot prove even its consistency restricted to a definable cut [Pudlák 1985] (Willard 2006a, Theorem 1, p. 4, `cited`). Solovay's theorem, communicated to Willard by telephone in April 1994 and never published by Solovay, states the conclusion (Willard 2006a, Theorem 2, p. 4, `cited`): no consistent system that proves successor total, together with the basic algebra of $+$ and $\times$, can prove its own Hilbert consistency. With subroutines, the price of faith is mere counting.

So the Hilbert-line self-justifying systems give up successor itself. They name numbers with constants $C_1, C_2, \ldots$, and the naming convention *is* the system (codified statement §5.5): incremental naming $C_i = C_{i-1} + 1$ works (ISREF(A), 2001, Theorem 3.4, p. 13, `full`); additive naming $C_i = C_{i-1} + C_{i-1}$ works and keeps reaching new numbers (ISCE(A), 2006a, Theorem 3, p. 6, `full`); multiplicative naming $C_i = C_{i-1} \cdot C_{i-1}$ fails (2006a, Theorem 4, p. 7, `full`). Remark 3, quoted in §3, explains why.

Other ways to buy faith

Adding the law of excluded middle as logical axioms puts the cut back into tableaux by the back door. The self-justifying system then becomes automatically inconsistent whenever its base proves the usual algebraic laws of $+$ and $\times$ (2020, Theorem 4.5, p. 13, `sketch`). Letting $\ddot{\mathsf{W}}$ say more about itself, namely consistency for a richer class of sentences (Willard's Level 2^+), brings the second theorem back for finite systems (2004, Theorem 1, p. 6, `full`).

[3] One nuance, to keep the boundary honest. What multiplication destroys is the flat consistency axiom and canonical reflection. Willard's ISM(A) recognizes multiplication as total and is still consistent, by what Willard calls a routine generalization of his proof; but its self-referential axiom is weaker, and it cannot prove the flat statement that no proof of $0 = 1$ exists (2001, §6; codified statement §7.2).

Proof system	Growth believed	$\ddot{\mathsf{W}}$ forbids	Outcome	Source, status
tableaux	+ total, × a relation	Ψ and $\neg\Psi$ (Π_1)	self-justifying	2005 Thms 3, 5, `full`
tableaux	× total	$0 = 1$	G2	2002a Thm 3.5 `full`; 6.4 `sketch`; 2007 Thm 6 `full`
tableaux + LEM axioms	+ total	Ψ and $\neg\Psi$	inconsistent	2020 Thm 4.5, `sketch`
tableaux	+ total	level 2^+	G2	2004 Thm 1, `full`
Hilbert	successor total	$0 = 1$	G2	Solovay, `cited`
Hilbert	additive names only	$0 = 1$	self-justifying	2006a Thm 3, `full`
Hilbert	multiplicative names	$0 = 1$	G2	2006a Thm 4, `full`

Read the table as Willard read it. The affirmative results are mostly `full`; the negative side contains sketches and one unpublished theorem. The boundary is sharply located in some coordinates and only bracketed in others (codified statement §8.2).

8 Trust Within Reach

A system *trusts* its proofs if it proves reflection: «if $\mathcal{F}$ proves that A never halts, then A never halts», for every Π_1 claim at once. That is too much to ask. No consistent system of this kind, proving the Π_1 theorems of (a version of) Peano Arithmetic, proves canonical reflection for every Π_1 sentence (2001, Theorem 7.2, p. 38, `full`); it is Löb's theorem in Willard's setting. What Willard's systems can prove is *tangibility reflection* (2001, Definition 1.2, Eq. 2):

$$\forall x \, \{ \, [\exists y \, \mathrm{Prf}_\alpha(\ulcorner \Psi \urcorner, y) \wedge \mathit{Tangible}(x)] \supset \Psi^x \, \},$$

where Ψ^x is Ψ with its unbounded universal quantifiers bounded by x. A number is tangible when it has room above it: some v with $x < v - 1$, or $x < v^{1/k}$. ISREF(A) proves the first form, given a condition on proof sizes (2001, Theorem 3.5, p. 14, `full`), and the tableaux system $\mathrm{IS}^\lambda(A)$ proves the second, under conditions on λ, k and A (Theorem 6.1, p. 34, and Corollary 6.2, p. 35, both `full`). In computational terms: *I trust my verdicts on every input for which there is memory to spare.* For a Σ_1 claim the bound costs nothing. With Ψ taken to be $0 = 1$, tangibility reflection gives back the flat statement that no proof of $0 = 1$ exists (2001, p. 8).

9 Taking Off the Glasses

Oberhoff's §5 takes the glasses off $\ddot{\mathsf{G}}$ and finds "I am not provable" underneath. Take them off $\ddot{\mathsf{W}}$ and you find "nothing refutes me", that is, "I am consistent". Over Peano Arithmetic, $\ddot{\mathsf{W}}$ is refuted, and $\mathsf{C}(\mathsf{C})$ finds the refutation: Theorem 6, written out. In Willard's systems that refutation would have to establish the existence of a refutation, a transcript-sized object, from inside a proof smaller than the object. The snake cannot swallow its tail, because its mouth is smaller than its body.

So self-reference is not where the trouble lies. The recursion theorem builds $\ddot{\mathsf{W}}$ in Willard's systems as easily as in Peano Arithmetic. What costs something is *self-simulation*: running yourself in your head and writing down the result. The second incompleteness theorem needs both. The first needs only self-reference and watching, which is why Willard's systems obey it.

Oberhoff found that his classical and computational proofs were "the exact same". The same holds here. Willard's arithmetic (a least Gödel number y, constants below $\sqrt{y}$, fewer than $\frac{1}{3}\log_2 y$ doublings) and the computational picture (a shortest refutation, the memory a proof occupies, a walk down the tree) are one argument. The arithmetic of Gödel numbers is the bookkeeping of memory.

10 Refactoring the Incompleteness Theorems, Again

Oberhoff's Theorem 7 is two theorems.

> **Theorem 7a** (Watching). *Every formal system capable of reasoning about elementary arithmetic proves, of each halting computation, that it halts.*
>
> **Theorem 7b** (Knowing that one watches). *Such a system proves Theorem 7a about itself.*

Theorem 7a is enough for the first incompleteness theorem in all of Oberhoff's forms. Theorem 7b is what Theorem 6 takes on faith, and it does not follow from 7a. Willard's systems satisfy 7a and prove their own consistency, so by Proposition E they fail 7b. Read in Willard's results, the cost of 7b depends on the proof system. Without lemmas, the second incompleteness theorem returns with uniform multiplication; with lemmas, it returns already with successor. In both cases what returns is not always 7b itself but enough of it, a compressed proof that a big enough number exists, for Gödel's argument to run.

So Gödel's work splits into three parts, not two. The first is programming the integers (7a). The second is teaching the integers that they have been programmed (7b). The third is the victory laps. Willard's self-justifying systems live in the gap between the first and the second. The second incompleteness theorem is a victory lap only for a runner who has already paid for 7b, and the receipt, in Willard's accounting, reads: multiplication if you never reuse a lemma, counting if you do.

Moral of the story: you may declare yourself a very stable genius, provided you cannot multiply. If you like to reuse your lemmas, you must also be unable to count.

11 What Self-Justification Does Not Buy

Willard was more careful about the limits of his results than most of his readers have been, and an honest retelling must carry his caveats.

- **A reduced question.** Willard (2016, Remark 4.9, p. 13) splits Hilbert's question in two: can a system verify its own consistency in a robust sense, and can it appreciate its consistency in some reduced but not immaterial sense? The answer to the first is no, by Gödel, Hilbert–Bernays and Pudlák–Solovay. Everything above answers the second.
- **A one-line proof.** A system that proves its consistency from its own axiom $\ddot{\mathsf{W}}$ has "essentially a 1-line proof" (2011, Remark 6.16b, p. 28), which is instinctive faith rather than proof-justification. The substantive justification lies outside: the theorem that *Probe* defeats every refutation, proved in the metatheory under the hypothesis that A's Π_1 theorems are true.
- **Almost cheating.** Willard raised the objection against himself: "Is it not almost cheating when an axiom system verifies its own consistency by using $\oplus$'s formalized 'I am consistent' axiom as an intermediate step, to verify its own consistency? After all, such a technique can verify its own consistency only in a technically purely legalistic sense" (2016, §8). His answer, *Platonic Stability*, is that a thinking being needs no sophisticated justification of its own reasoning, only a formalism in which presuming its own consistency provably will not spin it into inconsistency.
- **The first theorem stands.** Willard's systems are incomplete; they cannot decide $\ddot{\mathsf{G}}$ or $\ddot{\mathsf{R}}$.

- **Interpretation versus source.** The vocabulary of watching and knowing, the program C, the horizon picture and Proposition E are our reading, not Willard's formulations. His own words for the mechanism are "adequate memory" (1994) and the comparison, in 2006a Remark 3, between a proof's Gödel number and the integer the proof shows to exist.

(The cake promised in Oberhoff's abstract is still on its way. We cannot prove that it exists: it is one multiplication away.)

A Provenance

Page numbers refer to the author copies in the repository's `nachlass/papers/`; statuses are those of `nachlass/codification/registry/results.md`. "Verbatim, image" marks quotations checked in this session against rendered page images.

Paper	Item	Page	Content used here	Status
TR 93-10	—	6	philosopher's hedge; G1 but not G2; "either fails ... or fails to meet"	verbatim, image
TR 93-10	—	7	adequate memory; a_{300}; cuts and Solovay	verbatim, image
2001	Thm 3.4	13	ISREF(A) consistency-preserving	full
2001	Thm 3.5	14	ISREF(A) proves *TangPred* reflection (given Eq. 13)	full
2001	Lem 4.1	17	constants in y below $\sqrt{y}$	full
2001	Lem 4.2	19	parameters below y on addition-conservative branches	sketch
2001	Thm 4.3	20	IS(A) consistency-preserving	full
2001	Rem 4.4	20	Assertion ++, vacuous truth, $p \leq \mathrm{VALUE}(u) < y$	verbatim, image
2001	Rem 4.5	21	ISMULT has no analogue of Lem 4.2	—
2001	Lem 4.6, 4.8	22–23	affirmative branch cannot close; least refutation has one	full
2001	Case 11	28	"substantially more than six bits"	image (extraction record)
2001	Thm 6.1, Cor 6.2	34–35	$\mathit{TangRoot}_k$ reflection, Hilbert and tableaux	full
2001	Thm 7.2	38	no canonical Π_1 reflection	full
2002a	Thm 2.2	7	cut elimination super-exponential	—
2002a	Thm 3.5	11	$Q + V$: G2 for tableaux	full
2002a	Thm 6.4	25	$I\Sigma_0$: G2 for tableaux	sketch
2000	§5	—	"the true reason" (Lemma 8)	verbatim (codified §5.3)
2004	Thm 1	6	level 2^+ consistency: G2	full
2005	§1	—	Kleene, Rogers, Jeroslow	verbatim (codified §2.2)
2005	Def 5, Thm 1	19	θ-compactification; engine	full
2005	Rem 4	21	multiplication makes Def 5 unsatisfiable	—
2005	Lem 1	23	*Probe*, positive branch	full
2005	Thm 2	26	tableaux: $\theta = 1/4$	full
2005	Thms 3, 5	27, 29	$\mathrm{IS}_D(\cdot)$ consistency-preserving	full
2006a	Thms 1, 2	4	Pudlák's and Solovay's theorems	cited
2006a	Thms 3, 4	6–7	additive naming works; multiplicative fails	full
2006a	Rem 3	24	proof's Gödel number vs. the integer it proves exists	verbatim, image
2007	Thm 6	29	T-1, T-2: G2 for tableaux	full
2011	Rem 6.16b	28	"essentially a 1-line proof"	—
2016	Def 3.4, Ex 3.5	—	self-justification; "ironically typically useless"	verbatim (codified §2.2)
2016	Rem 4.9; §8	13; —	the two questions; "almost cheating"	verbatim (codified §1.4)
2020	Ex 3.1; Thm 4.5	13	apparatus identity; Xtab inconsistent	sketch (Thm 4.5)

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